

Additional file 1

Additional file 2

Additional file 3

Additional file 4

Additional file 5

Additional file 6

Additional file 7

Additional file 8

Additional file 9

Additional file 10

Additional file 11

Additional file 12

Default hyperparameters for all models evaluated in this study. Default hyperparameter settings used for every model architecture on the QM9 HOMO–LUMO gap experiments.

Model	Hyperparameter	Default Value
<i>Tree-based Models</i>		
Random Forest	n_estimators	100
	max_depth	None
	max_features	sqrt
	min_samples_leaf	1
	min_samples_split	2
	bootstrap	True
Quantile Forest	n_estimators	300
	max_depth	None
	max_features	sqrt
	min_samples_leaf	1
	min_samples_split	2
	bootstrap	True
XGBoost	quantile	0.5
	n_estimators	100
	max_depth	6
	learning_rate	0.1
	subsample	1.0
	colsample_bytree	1.0
	colsample_bylevel	1.0
	min_child_weight	1
	gamma	0.0
	reg_alpha	0.0
	reg_lambda	1.0
LightGBM	n_estimators	100
	learning_rate	0.1
	num_leaves	31
	max_depth	−1 (unlimited)
	subsample	1.0
	colsample_bytree	1.0
	min_child_samples	20
	reg_alpha	0.0
	reg_lambda	0.0
NGBoost	learning_rate	0.01
	n_estimators	500
	natural_gradient	True
<i>Kernel-based Models</i>		
SVM	C	1.0
	gamma	scale
	kernel	rbf
Gauche (GP)	kernel	Tanimoto
	outputscale	1.0
	likelihood_noise	0.001
<i>Neural Networks</i>		
NN- α	hidden_size_1	128

Pairwise Spearman rank correlations between model noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap ($\rho \geq 0.95$). “Model A” and “Model B” label each pair, both treated symmetrically. Checkmarks indicate pairs where one model was excluded from the ANOVA due to near-redundancy. Gauche (RBF) pairs omitted (all $\rho = 1.0$, $n = 6$, PDV-only artifact).

Model A	Model B	n	ρ	Excluded
VBLL- β	NGBoost	18	0.998	
LightGBM	XGBoost	46	0.998	
Gauche	VBLL- β	18	0.996	✓
QRF	RF	40	0.995	✓
VBLL- α	LightGBM	24	0.992	
RF	XGBoost	46	0.989	
NGBoost	XGBoost	40	0.988	
NGBoost	RF	40	0.987	
LightGBM	RF	46	0.986	
VBLL- β	SVM	24	0.986	
QRF	XGBoost	40	0.986	✓
NGBoost	QRF	34	0.986	✓
VBLL- α	NGBoost	18	0.986	
VBLL- α	XGBoost	24	0.984	
LightGBM	NGBoost	40	0.984	
LightGBM	QRF	40	0.984	✓
VBLL- α	VBLL- β	24	0.983	
NN- α	NN- β	52	0.983	
LightGBM	SVM	46	0.982	
BNN- β	SVM	34	0.981	
VBLL- α	SVM	24	0.979	
BNN- α	VBLL- β	24	0.979	
VBLL- α	QRF	24	0.978	✓
SVM	XGBoost	46	0.978	
BNN- α	BNN- β	36	0.977	
BNN- α	SVM	34	0.976	
VBLL- α	Gauche	18	0.975	✓
VBLL- α	RF	24	0.975	
QRF	SVM	40	0.974	✓
BNN- α	VBLL- α	24	0.974	
LightGBM	VBLL- β	24	0.973	
VBLL- α	BNN- β	24	0.972	
Gauche	SVM	40	0.972	✓
Gauche	XGBoost	44	0.970	✓
NGBoost	SVM	40	0.970	
NN- α	BNN- β	36	0.970	
Gauche	LightGBM	42	0.969	✓
RF	SVM	46	0.969	
BNN- β	VBLL- β	24	0.968	
VBLL- β	QRF	24	0.966	✓
NN- α	BNN- α	36	0.965	
VBLL- β	XGBoost	24	0.965	
NN- β	SVM	46	0.962	
NN- α	SVM	46	0.960	
Gauche	NGBoost	34	0.959	✓
LightGBM	BNN- β	36	0.959	
BNN- α	NN- β	36	0.958	
BNN- α	QRF	28	0.956	✓
Gauche	QRF	34	0.956	✓
VBLL- β	RF	24	0.955	
Gauche	RF	40	0.954	✓
BNN- β	XGBoost	34	0.953	
BNN- β	QRF	28	0.952	✓
NN- α	VBLL- β	24	0.952	
NN- β	BNN- β	36	0.952	
BNN- α	LightGBM	36	0.951	
NN- β	VBLL- β	24	0.950	
BNN- α	Gauche	30	0.950	✓

Pairwise Spearman rank correlations between representation noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap. “Rep A” and “Rep B” label each pair of molecular representations, both treated symmetrically. Checkmarks indicate pairs where one representation was excluded from the ANOVA.

Rep. A	Rep. B	n	ρ	Excluded
ECFP4	Morgan	60	0.995	✓
MHGGNN	Morgan	48	0.994	✓
Rand. SMILES	SMILES	16	0.994	✓
Rand. SMILES	SNS	16	0.994	✓
ECFP4	MHGGNN	54	0.993	
ECFP4	SNS	56	0.992	✓
Morgan	SNS	52	0.992	✓
MHGGNN	SNS	44	0.991	✓
PDV	ECFP4	72	0.990	
PDV	MHGGNN	48	0.989	
SMILES	SNS	52	0.988	✓
Morgan	Rand. SMILES	16	0.988	✓
Morgan	SMILES	54	0.988	✓
PDV	Morgan	54	0.986	✓
PDV	Rand. SMILES	12	0.986	✓
PDV	SNS	50	0.984	✓
ECFP4	SMILES	72	0.984	
Mol2Vec	Morgan	48	0.982	✓
MHGGNN	Mol2Vec	54	0.982	
ECFP4	Mol2Vec	54	0.982	
PDV	PDV (binary)	72	0.980	✓
MHGGNN	SMILES	48	0.980	
PDV	Mol2Vec	48	0.979	
ECFP4	Rand. SMILES	16	0.979	✓
PDV	SMILES	66	0.978	
MHGGNN	PDV (binary)	54	0.972	✓
Mol2Vec	SNS	44	0.969	✓
Mol2Vec	SMILES	48	0.966	
ECFP4	PDV (binary)	78	0.965	✓
Morgan	PDV (binary)	60	0.962	✓
PDV (binary)	SNS	56	0.961	✓
Mol2Vec	PDV (binary)	54	0.956	✓
MHGGNN	Rand. SMILES	16	0.956	✓
PDV (binary)	SMILES	72	0.941	✓
PDV (binary)	Rand. SMILES	16	0.929	✓
Mol2Vec	Rand. SMILES	16	0.891	✓

Intraclass correlation coefficients (ICC(1,1)) for model pairs with $\text{ICC} \geq 0.5$, assessing within-family consistency of noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap. ICC(1,1) measures the proportion of total variance attributable to between-configuration differences; values near 1.0 indicate that two models rank configurations almost identically. The Family column groups each pair: “Tree” (both tree-based), “ α/β ” (architecture-pair partners such as BNN- α /BNN- β), “ α ” (both share the α neural-network backbone), or “Cross” (cross-family pair).

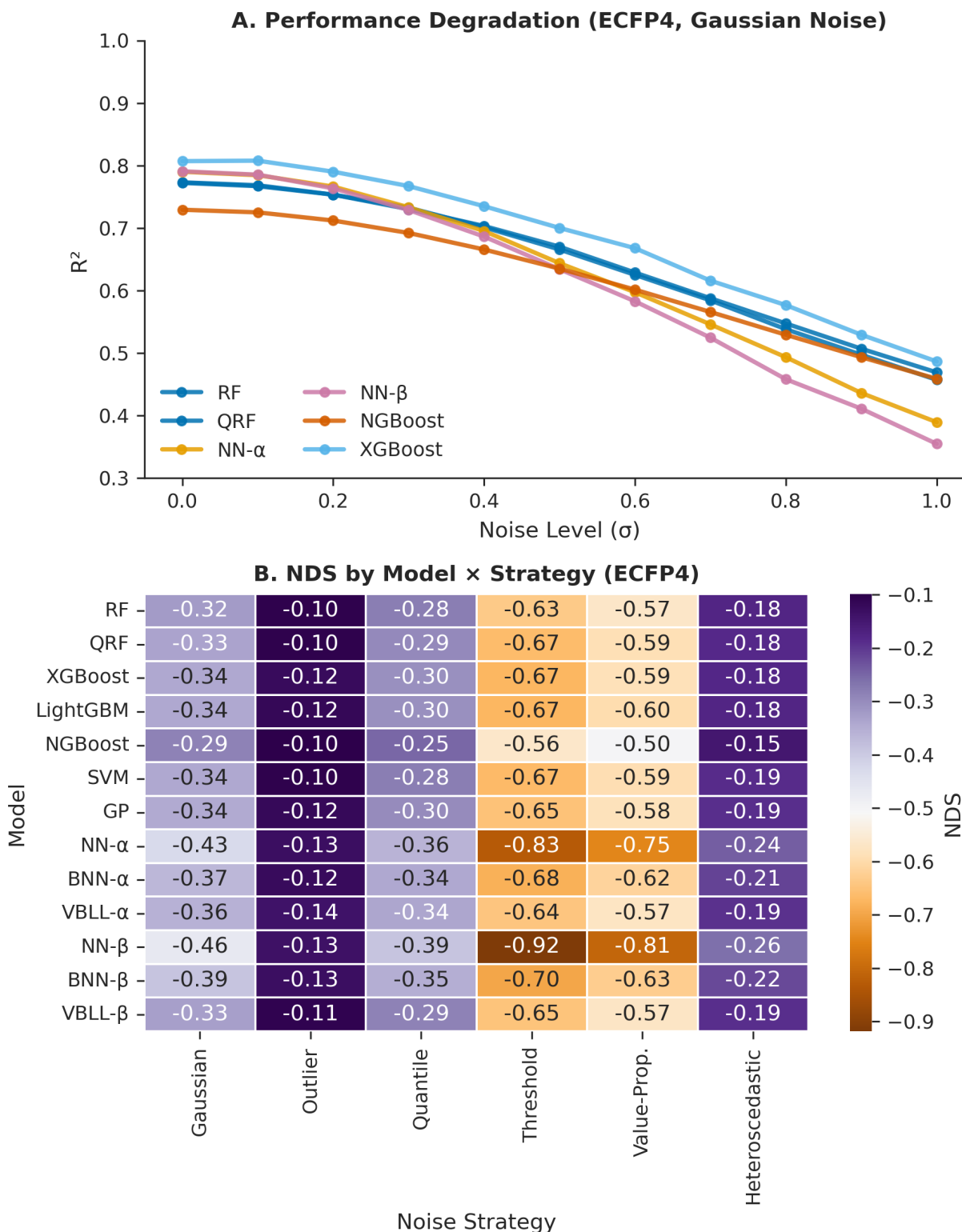
Model A	Model B	n	ICC(1,1)	Mean $ \Delta\text{NDS} $	Family
LightGBM	XGBoost	8	0.944	0.0060	Tree
BNN- α	BNN- β	6	0.846	0.0101	α/β
QRF	RF	7	0.810	0.0216	Tree
VBLL- α	VBLL- β	4	0.782	0.0108	α/β
QRF	XGBoost	7	0.754	0.0157	Tree
LightGBM	QRF	7	0.743	0.0133	Tree
VBLL- α	LightGBM	4	0.698	0.0115	Cross
VBLL- β	SVM	4	0.689	0.0133	Cross
RF	XGBoost	8	0.678	0.0212	Tree
VBLL- α	XGBoost	4	0.656	0.0133	Cross
BNN- α	VBLL- α	4	0.604	0.0134	α
LightGBM	RF	8	0.586	0.0244	Tree
LightGBM	VBLL- β	4	0.577	0.0174	Cross
LightGBM	SVM	8	0.561	0.0117	Cross
BNN- α	VBLL- β	4	0.552	0.0123	α/β
VBLL- β	XGBoost	4	0.544	0.0195	Cross
VBLL- α	QRF	4	0.540	0.0229	Cross
VBLL- α	SVM	4	0.522	0.0124	Cross
Gauche	VBLL- β	3	0.519	0.0113	Cross

Configurations excluded from robustness analysis due to baseline $\text{R}^2 \leq 0.6$ on the QM9 HOMO–LUMO gap, summarized by model and representation. Each cell shows the number of noise strategies (out of 6) for which the configuration was excluded.

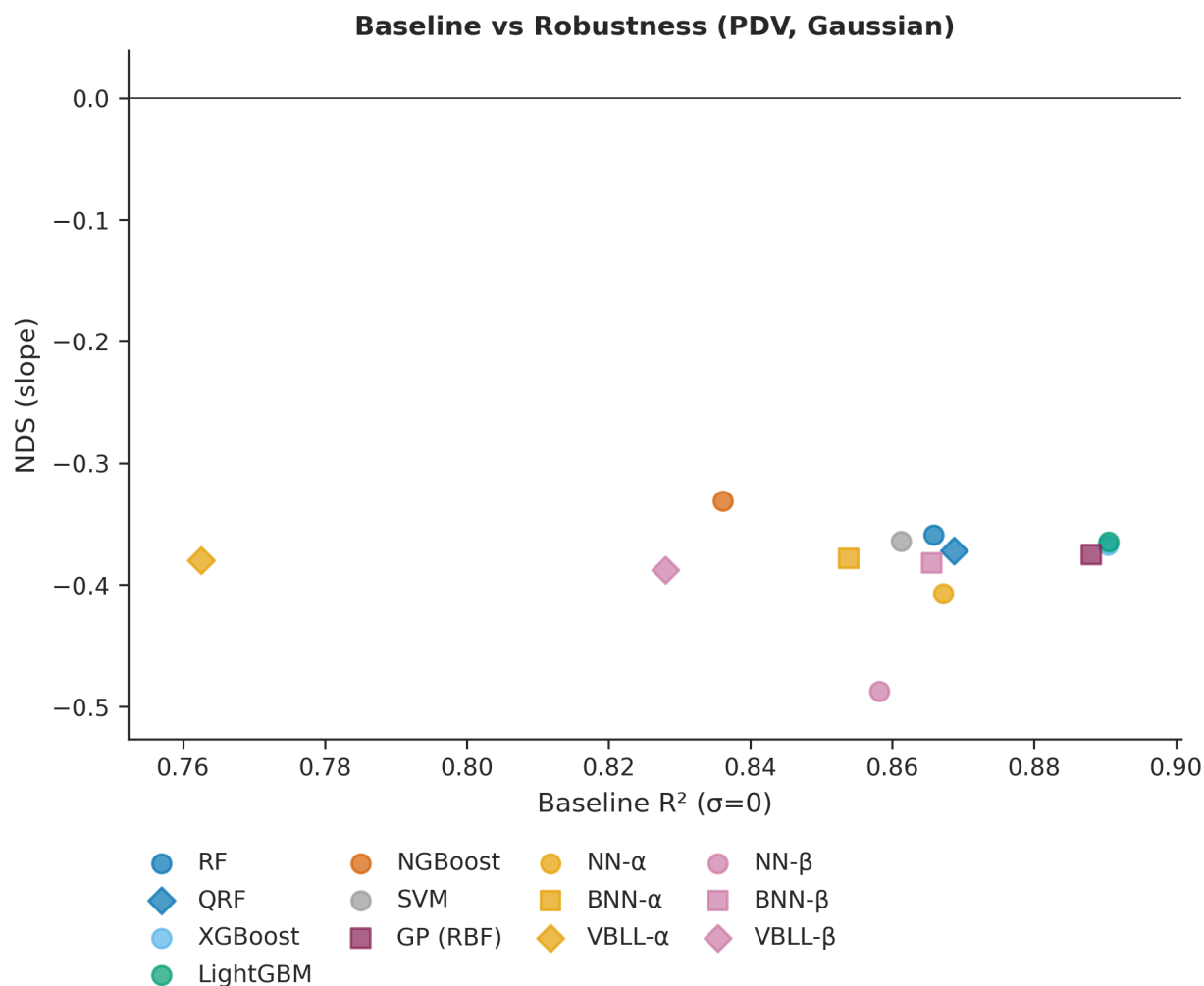
Model	Mol2Vec	MHGGNN	SMILES	Rand. SMILES
BNN- α	6	6	—	—
BNN- β	6	6	—	—
VBLL- α	6	6	—	—
VBLL- β	6	6	—	—
RF	—	—	—	4
QRF	—	—	—	4
NGBoost	—	—	6	4

Neural network Bayesian transformation effects on noise degradation slope (NDS) by representation, on the QM9 HOMO–LUMO gap. Positive Δ NDS indicates the Bayesian variant is more robust (less negative slope) than its deterministic counterpart.

Model	Representation	Baseline R^2	NN- β NDS	BNN- β NDS	Δ NDS
NN- β	ECFP4	0.791	−0.464	−0.388	+0.076
NN- β	PDV	0.858	−0.487	−0.382	+0.106
NN- β	SMILES	0.721	−0.558	−0.365	+0.193



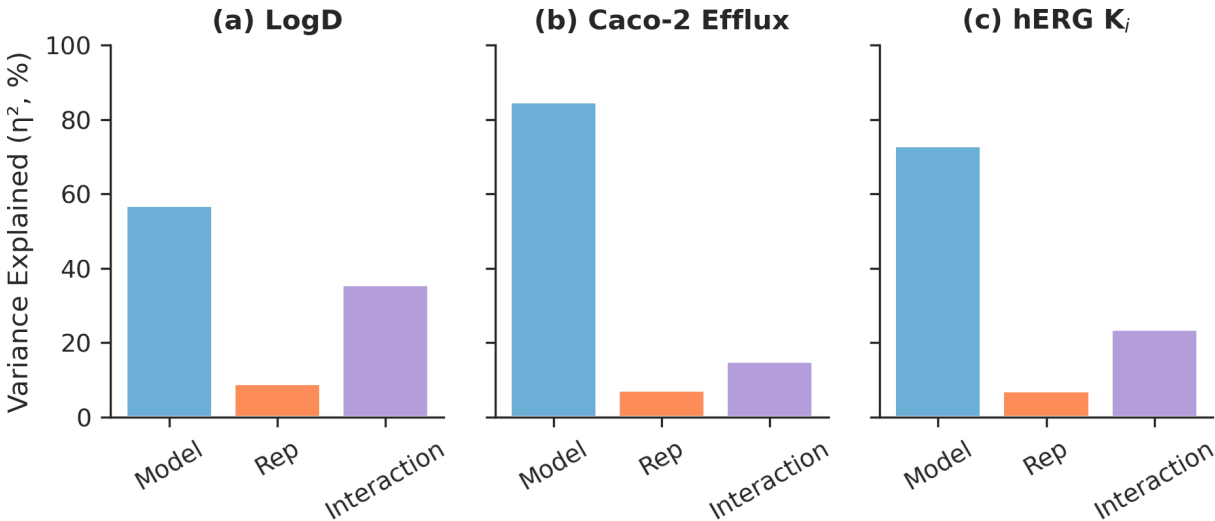
Global overview of noise robustness on the ECFP4 representation, evaluated on QM9 HOMO–LUMO gap. (A) Performance degradation curves (R^2 versus noise level σ) under Gaussian noise. (B) Noise degradation slope (NDS) heatmap across model architectures and noise strategies; lower (more negative) NDS indicates faster degradation under noise. **Takeaway:** the same model-level ranking observed on PDV in the main paper (Figure 4) is reproduced on ECFP4 — NGBoost and SVM remain most robust, NN- β degrades fastest — confirming that the robustness ordering is a property of model architecture rather than the underlying feature space.



Baseline R^2 versus noise degradation slope (NDS) scatter for all models on the physicochemical descriptor vector (PDV) representation under Gaussian noise, on the QM9 HOMO–LUMO gap. Lower (more negative) NDS indicates faster performance loss under noise. **Takeaway:** NDS sits near -0.38 for nearly all models regardless of their baseline R^2 (which spans 0.76 – 0.89), so a high-performing model on clean data is not automatically a more robust one — clean-data benchmarks alone are not a reliable proxy for noise robustness.

Uncertainty quantification metrics for probabilistic models on the ECFP4 representation (Gaussian noise strategy) on the QM9 HOMO–LUMO gap. Unc-Error ρ : Spearman correlation between predicted uncertainty and absolute prediction error. Unc-Noise ρ : Spearman correlation between predicted uncertainty and injected noise level σ . ECE: expected calibration error (lower = better-calibrated). Coverage targets: 68% (1σ), 95% (2σ).

Model	Unc-Error ρ	Unc-Noise ρ	ECE	Cov. 1σ	Cov. 2σ
QRF	0.25	0.27	0.19	74.7%	94.6%
NGBoost	0.22	0.37	0.11	69.2%	94.2%
GP	0.18	0.34	0.23	78.6%	96.0%
BNN- β	0.20	0.37	0.29	78.6%	95.2%
BNN- α	0.15	0.32	0.24	76.6%	95.1%
VBLL- β	0.19	0.34	0.19	73.9%	95.5%
VBLL- α	0.11	0.31	0.22	74.5%	95.1%



ANOVA variance decomposition (η^2 , %) on external validation datasets: (a) LogD, (b) Caco-2 Efflux, and (c) hERG K_i . Model, representation, and interaction contributions to noise degradation slope (NDS) variance for each dataset; lower (more negative) NDS indicates faster performance loss under noise. **Takeaway:** model architecture is the single largest source of NDS variance on every external dataset (57–85%, vs $\leq 10\%$ for representation), echoing the QM9 finding in the main paper (Figure 2B) — robustness on small experimental ADME data is still primarily a function of model choice.

Random forest (RF) vs quantile regression forest (QRF) noise degradation slope (NDS) on the three external validation datasets (LogD, Caco-2 Efflux, hERG K_i). QRF is consistently less robust than RF across all external datasets.

Dataset	RF NDS	QRF NDS	Δ NDS
ChEMBL-hERG-Ki	−0.231	−0.359	−0.128
OpenADMET-Caco2	−0.534	−0.627	−0.094
OpenADMET-LogD	−0.101	−0.185	−0.083

Default hyperparameters used for external validation experiments on LogD, Caco-2 Efflux, and hERG K_i . All models used standard defaults except SVM, which used representation-specific tuned kernels to maintain consistency across representations.

Model	Hyperparameters	Source
RF	$n_{\text{estimators}} = 100$	Default
QRF	$n_{\text{estimators}} = 100$	Default
XGBoost	$n_{\text{estimators}} = 100$	Default
LightGBM	$n_{\text{estimators}} = 100$	Default
NGBoost	$n_{\text{estimators}} = 500$, $\text{lr} = 0.01$	Default
NN- α	$[128, 64]$, dropout = 0.2, 100 epochs	Default
GP	RBF + constant + white noise kernel, $N_{\text{max}} = 2000$	Default
SVM (ECFP4)	RBF kernel, $C = 1.42$, $\gamma = \text{scale}$	Tuned
SVM (PDV)	Poly kernel (degree 2), $C = 2.04$, $\text{coef}_0 = 1.91$	Tuned
SVM (SNS)	Poly kernel (degree 2), $C = 1.08$, $\text{coef}_0 = 2.71$	Tuned
SVM (MHG-GNN)	RBF kernel, $C = 1.0$, $\gamma = \text{scale}$	Default